

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
Examination Control Division
2082 Bhadra

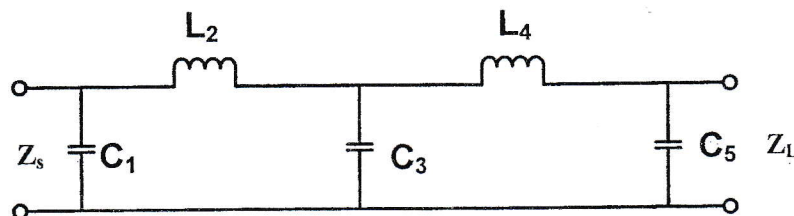
Exam.	Regular		
Level	BE	Full Marks	80
Programme	BEI	Pass Marks	32
Year / Part	IV / I	Time	3 hrs.

Subject: - RF and Microwave Engineering (EX 716)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Necessary formulas and Smith Charts are attached herewith.
- ✓ Assume suitable data if necessary.

1. Compare the behavior of microwave passive components at low frequency bands against microwave bands. Prepare the detail real-world protocols of any three major applications of microwave communication systems. [4+6]
2. A broadband microstrip antenna with a load exhibiting reflection coefficient of $0.33 \angle 66^\circ$ is connected to a transmission patch having impedance of 75Ω . Design the appropriate matching stubs. Express the appropriate scattering matrix of your designed matched networks. [10]
3. Design a model of a three-port network and define its characteristic parameters. Prepare the S-Matrix of a rectangular magic tee having shorted both E and H-arms and explain how it behaves. [4+4]
4. Prove that the lowest cut-off frequency mode in rectangular waveguides operating in TM mode is TM_{11} . Explain the working principle of any one of circular beam-type microwave cavity-controlled cross-field amplifier. [3+7]
5. Find the basic differences of the systems having following sets S-Matrix. [4]
 - a) $S_{11} = 0.65 \angle 146^\circ$, $S_{12} = 0.12 \angle 46^\circ$, $S_{21} = 2.30 \angle 44^\circ$, $S_{22} = 0.17 \angle -172^\circ$
 - b) $S_{11} = 0.30 \angle -134^\circ$, $S_{12} = 0.03 \angle 46^\circ$, $S_{21} = 2.20 \angle 44^\circ$, $S_{22} = 0.10 \angle -172^\circ$

Using the given S-Parameters given in (a), determine the stability and compare maximum power gains for unilateral and bilateral modes using supplied formulas. [12]
6. Explain, why microwave filters are designed using the Insertion Loss Method? Explain the detail design steps of the following microwave filter, and implement the circuit using microstrip. [3+8+3]



7. Describe how microwaves are hazardous to humans. Explain the working principle of a microwave measuring device to map different radiation zones around a GSM BTS. [4+8]

Supplied Formulas:

$$K = \frac{1 + |\Delta|^2 - |S_{11}|^2 - |S_{22}|^2}{2|S_{12}||S_{21}|}$$

$$\Delta = \frac{(S_{11}S_{22} - S_{12}S_{21})}{1 - |S_{11}|^2}$$

$$\mu = \frac{|S_{22} - \Delta S_{11}^*| + |S_{21}S_{12}|}{|S_{22} - \Delta S_{11}^*| + |S_{21}S_{12}|}$$

$$\Gamma_S = \frac{B_1 \pm \sqrt{B_1^2 - 4|C_1|^2}}{2C_1}$$

$$\Gamma_L = \frac{B_2 \pm \sqrt{B_2^2 - 4|C_2|^2}}{2C_2}$$

$$\text{Where, } B_1 = 1 + |S_{11}|^2 - |S_{22}|^2 - |\Delta|^2,$$

$$B_2 = 1 + |S_{22}|^2 - |S_{11}|^2 - |\Delta|^2$$

$$C_1 = S_{11} - \Delta S_{22}^*$$

$$C_2 = S_{22} - \Delta S_{11}^*$$

$$C_L = \frac{(S_{22} - \Delta S_{11}^*)}{|S_{22}|^2 - |\Delta|^2}$$

$$C_S = \frac{(S_{11} - \Delta S_{22}^*)}{|S_{11}|^2 - |\Delta|^2}$$

$$R_L = \frac{|S_{12}S_{21}|}{|S_{22}|^2 - |\Delta|^2}$$

$$R_S = \frac{|S_{12}S_{21}|}{|S_{11}|^2 - |\Delta|^2}$$

$$G_{Tmax} = \left(\frac{1}{1 - |\Gamma_S|^2} \right) |S_{21}|^2 \left(\frac{1 - |\Gamma_L|^2}{|1 - S_{22}\Gamma_L|^2} \right)$$

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Subject: - RF and Microwave Engineering (EX 716)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
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1. Find the unique differences between microwave and conventional low frequency systems. Mention detail specifications/protocols of any five microwave communication system. [4+4]
2. Compare radiation fields around a mobile BTS operating at 900 MHz and 1800 MHz frequency bands with their quarter-wavelength antennas. Explain how microwave EMR becomes hazardous to humans; and mention its IEEE safety standard. [4+2+2]
3. Write the electromagnetic field equations of a circular waveguide operating at TE Mode. Explain degenerative and dominant modes of a rectangular waveguide with suitable examples. [7+3]
4. Using proper S-Matrices, mention the properties of a Hybrid-Tee in the situation of (a) terminated E-arm, (b) terminated H-arm and (c) terminated co-planner arms. [4+4+2]
5. Design a double-stub matching network for an antenna operating at 10 GHz, having $\Gamma = |0.45| \angle 60^\circ$ and connected to a 100 Ohm transmission line. Sketch its physical diagram using micro strips. Prepare its S-Matrices before and after the design. [10+2+2]
6. Using following S-parameters: $S_{11}=0.55 \angle 150^\circ$, $S_{12}=0.04 \angle 20^\circ$, $S_{21}=2.82 \angle 180^\circ$ and $S_{22}=0.45 \angle 30^\circ$, design a microwave amplifier for maximum power gain for both bilateral and unilateral cases. [8+4]
7. Mention the details of microwave filter designing steps. Provide a sketch of triple-pad series-arm type HPF using micro strips. [7+3]
8. Write short notes on: (Any Two) [2×4]
 - a) Multi-cavity Klystron Amplifier
 - b) Circulating Calorimetry
 - c) Microwave Amplifier Design Flow and Stability Analysis

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1. Based on their respective operating principles, distinguish between microwave and traditional low frequency systems. Explain why S-parameter based analysis is a must in microwave network analysis. [5+3]
2. A 100Ω lossless transmission line is terminated with a complex load of $120-j160\Omega$. Design a single matching stub. Mention all the design steps with proper reasoning. Provide the S-Matrix for both before and after the design. [5+3+2]
3. a) Show that the dominant mode in rectangular waveguides operating in TE mode is TE_{10} , when $b > a$. [3]
b) Identify and explain, along with a neat diagram, the properties of the passive microwave device described by the following S-Matrix: [5]

$$S = \begin{bmatrix} S_{11} & S_{12} & S_{13} & S_{14} \\ S_{12} & S_{22} & S_{13} & -S_{14} \\ S_{13} & S_{13} & 0 & 0 \\ S_{14} & -S_{14} & 0 & 0 \end{bmatrix}$$

4. Express various field components of a rectangular waveguide in TM mode. [8]
5. Explain what are bunching effect and field effect. Describe the working principle of any one microwave high power oscillator, that employs bunching and field effects, with neat diagram. [2+2+4]
6. Investigate with neat flow diagram the stability of a transistor having following S-parameters: [3+8]
 $[S] = \begin{bmatrix} 0.6\angle -140^\circ & 0.03\angle 62^\circ \\ 2.40\angle 54^\circ & 0.70\angle -58^\circ \end{bmatrix}$
7. Illustrate an example of a single section series arm passive BPF using microstrip. How are microwave filters designed using Insertion Loss Method? [2+8]
8. Explain how microwave medium power is measure with the static and dynamic circulating calorimetry. List all the microwave measurables and their respective measuring tools. [6+3]
9. Approximate the hazardous and safe zone for BTS tower operating at 2600MHz with antenna size of 0.5 m. Explain, how non-radiating radiation is hazardous to the human body? [4+4]

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Subject - RF and Microwave Engineering (EX 716)

- Candidates are required to give their answers in their own words as far as practicable.
 - Answer All questions.
 - The figures in the margin indicate Full Marks.
 - Necessary Equations and Smith Charts are herewith.
 - Assume suitable data if necessary.
1. List out the uses of microwaves in communication systems and provide brief communication protocols of any three of them. [2+6]
 2. Classify microwave propagation fields of a GSM BTS and its radiation parameters. Explain what non-ionizing SAR is and how it becomes hazardous to human body. [4+4]
 3. Prepare S-Matrices of a perfectly working E-Plane Tee, H-Plane Tee and a Duplexer, and explore their uses. [3]
 4. Define the field equations and the basic characteristic equations of a rectangular waveguide working on TM_{11} mode. [8]
 5. Design a double-stub impedance matching network for a load of $\Gamma = 0.64 \angle 58^\circ$ connected to a 100Ω transmission line. Prepare the S-Matrix of its matched network. [8+2]
 6. Calculate the maximum gains of both bilateral and unilateral models of a transistor having the S-parameters of $S_{11} = 0.55 \angle -150^\circ$, $S_{12} = 0.04 \angle 20^\circ$, $S_{21} = 2.82 \angle 180^\circ$ and $S_{22} = 0.45 \angle -30^\circ$. Sketch a flow chart that describes the microwave amplifier designing procedure. [6+3+3]
 7. Explain the steps of microwave filter designing. Provide a sketch of double pad T-Type passive filter using a micro-strip. [7+3]
 8. Explain the working principles of a cross-field magnetron having 60° of phase differences among the adjacent cavities. [8]
 9. Write short notes on: (Any Two) [2×4]
- a) Stability analysis of a microwave amplifier
 - b) Double-channel Bolometer
 - c) Sketch of an Impedance Chart

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Subject: - RF and Microwave Engineering (EX 716)

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- ✓ Attempt All questions.
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- ✓ Necessary formulas and smith charts are attached herewith.
- ✓ Assume suitable data if necessary.



1. Differentiate between lumped and distributed circuit analysis. List out the areas of application of microwave system. Provide detail protocols of two microwave communication systems. [3+3+4]
2. A 75Ω coaxial line is terminated with a complex load of $109 + j120 \Omega$. Design a double-stub matching system using short-circuited coaxial lines. Prepare its S-matrix. [8+2]
3. Briefly explain the need of S-parameter and its properties in microwave analysis. Explain three port network parameters using an example of E-plane Tee. [4+4]
4. What is dominant mode in a waveguide? What are the dominant modes for rectangular and circular waveguides? Does a rectangular waveguide of width 2.254 cm support the propagation of a signal having frequency 6 GHz? [1+1+2]
5. Draw a neat diagram of a magic tee and derive its S-parameter. [4]
6. What is bunching effect? Briefly describe the construction and operation of Traveling Wave Tube (TWT). [2+6]
7. A GaAs MESFET has the following S-parameters measured with a 50Ω resistance at 10 GHz:
 $S_{11} = 0.55 \angle -160^\circ$ $S_{12} = 0.04 \angle 10^\circ$
 $S_{21} = 4.82 \angle 180^\circ$ $S_{22} = 0.45 \angle -30^\circ$ [10]
Determine the stability and compare maximum power gains for unilateral and bilateral modes using supplied formulas.
8. How is a low pass filter is prototyped based on Butterworth or Chebyshev approximations and converted into other types of filters? Implement a second order high pass filter π -section using microstrips. [10]
9. Briefly describe power measurement and the working principle of double channel Bolometer Bridge method. [8]
10. Write short notes on: (Any Two) [2×4]
 - a) HERP
 - b) TE_{mn} Field Equations of rectangular waveguide
 - c) Spectrum Analyzer

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1. How is microwave frequency band classified by the IEEE? Enumerate the basic advantages and disadvantages of microwaves compared to lower frequencies. [2+4]
2. Design a double stub shunt tuner to match a load impedance of $Z_L = 60 - j80\Omega$ to a 50Ω line. The stubs are to be open circuited stubs and are spaced $3\lambda/8$ apart. Assume the first is 0.4λ from the load. Formulate the S-matrix for your design. [8+2]
3. Why are S-parameters used in microwave network analysis? Derive the S-parameters for a two port network. [2+6]
4. For an air-filled rectangular waveguide with a width of 3 cm and a desired frequency of operation of 6 GHz (for dominant mode), determine cut-off frequency, cut-off wavelength, group velocity, phase velocity, propagation wavelength in the waveguide and the characteristic impedance. Explain the four basic parameters used to describe the performance of a directional coupler. [6+4]
5. Sketch a cross-sectional view of a magnetron having 45 degrees of phase shifts among the adjacent cavities, and explain its functioning as a power amplifier. Explain the bunching effect. [6+2]
6. Investigate the stability of a transistor having following S-parameters at 6GHz. [5+5]

$$[S] = \begin{bmatrix} 0.894\angle -60.6^\circ & 0.020\angle 62.4^\circ \\ 3.122\angle 123.6^\circ & 0.781\angle -27.6^\circ \end{bmatrix}$$
7. Explain why insertion loss technique is used to design microwave filters. With proper labeling sketch microwave double-section shunt – arm types microwave LPF using micro strips. [3+5]
8. Discuss the possible RF radiation fields to the public exposer. Explain, what are the international standards and recommended (SARPs) practices to safe from such radiation. [3+3]
9. Explain power measurement using calorimeter wattmeter. What are the limitations of using single bridge bolometer? [6+2]
10. Write short notes on: [2×3]
 - a) MASER
 - b) E-plane Tee

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Year / Part	IV / I	Time	3 hrs.

Subject: - RF and Microwave Engineering (EX 716)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
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- Compare the behavior of circuits for Low frequency/ Conventional and RF/ Microwave bands. Classify microwave frequency band and its application in major areas. [3+3]
- A 50Ω lossless transmission line is required to be matched with the load admittance $0.00813 + j0.0065 \Omega$, by a double-stub shunt tuner with separation of $3\lambda/8$ and the distance of the first stub from the load is 0.01λ . Calculate the length of each stub by using the smith chart. Write the s-parameter for the matched network. [8+2]
- Why S-parameters are used in high frequencies? The S-matrix of certain microwave network is given as

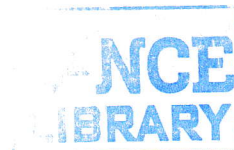
$$S = \begin{bmatrix} 0.4 + j0.5 & j0.6 \\ j0.6 & 0.4 - j0.5 \end{bmatrix}$$
 [3+1+1+1+2]
 - Is the network reciprocal?
 - Is the network lossless?
 - What is the return loss at the input?
 - If the input power to the network is 5 watts. What is the reflected power?
- Provide the fundamental field and characteristic equations of a circular waveguide for TE mode. [10]
- How is the output of conventional tubes reduced at microwaves due to inter-electrode capacitance, lead inductance and transit time effect? Explain about the construction and working principle of TWT. [2+6]
- For transistor having following S-parameter $S_{11} = 0.894 \angle -60.6^\circ$, $S_{21} = 3.122 \angle 123.6^\circ$, $S_{12} = 0.020 \angle 62.4^\circ$, $S_{22} = 0.781 \angle -27.6^\circ$. Determine the stability and compare maximum power gains for bilateral and unilateral modes. [5+5]
- How is a low pass filter prototype based on Butterworth approximation designed using insertion loss method? Implement a low pass filter π section using microstrips. [6+2]
- Explain the RF/MW radiation hazards and its safety practices. [3+3]
- List out the major RF/MW measurement parameters. How the VSWR of any microwave transmitter (In case of $VSWR > 10$) can be measured? Explain. [2+6]
- Write short notes on: [2×3]
 - Microwave Magic Tee
 - Gunn-diode

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Year / Part	IV / II	Time	3 hrs.

Subject: - RF and Microwave Engineering (EX 752)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Necessary formula and Smith chart are attached herewith.
- ✓ Assume suitable data if necessary.



1. Explain how microwave systems are different from conventional low frequency devices. [8]
2. Define SAR and discuss microwave radiation hazards based on the SAR exposures. [10]
3. Define EM field equations and its characteristic parameters of a rectangular waveguide working on TE mode. [10]
4. Compare h-parameters with S-parameter. Using a two-port network describe S-parameters. [5+5]
5. Assuming a load of $110 + j110$ ohm is connected to a 100-ohm transmission line, find the lengths and spacing for a double-stub impedance matching system using three-eighths wavelength separation between the stubs. [12]
6. Check the stability and find the unilateral gain of a transistor having following S-parameters: $S_{11}=0.45\angle-160^\circ$, $S_{12}=0.04\angle20^\circ$, $S_{21}=3.12\angle180^\circ$ and $S_{22}=0.35\angle-40^\circ$. [7+8]
7. Write short notes (Any Three) [3×5]
 - a) Microwave magic tee
 - b) Bolometry
 - c) Multi-cavity klystron
 - d) Immittance Chart
 - e) Spectrum Analyzer

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Subject: - RF and Microwave Engineering (EX 752)

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- ✓ Necessary Equations and Smith Charts are attached herewith.
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1. Differentiate the behavior of a transmission line at conventional low frequency and microwave bands. [10]
2. List and describe five major RF safety practices. Differentiate among microwave reactive, near - and far- field radiations with necessary illustrations. [5+5]
3. Explain the working principle of bolometry used in microwave measurements. [10]
4. A lossless 50 Ohm transmission line is terminated by an impedance of $75 + j100$ Ohm. Using Smith Chart, find (a) Γ_L , (b) VSWR, (c) Z_{in} at a distance of 0.375λ from the load, (d) the shortest length of line for which impedance is purely resistive, and (e) the value of this resistance. [10]
5. With self -defined line and load impedances and mentioning the steps, design a single short- circuited matching stub. [10]
6. A GaAs power FET has the S – parameters at 5 GHz with 50 – ohm line measured as $S_{11} = 0.45 \angle 163^\circ$, $S_{12} = 0.04 \angle 40^\circ$, $S_{21} = 2.55 \angle -106^\circ$ and $S_{22} = 0.46 \angle -65^\circ$. Using these parameters, design an amplifier and find the total gain power gain. [10]
7. Write the basic field equations of a rectangular waveguide working at TE mode. [10]
8. Write short notes on (Any Two) [2×5]
 - a) Magnetron
 - b) Directional coupler
 - c) Magic Tee
 - d) S-parameters

Supplied Formulas:

$$K = \frac{1 + |\Delta|^2 - |S_{11}|^2 - |S_{22}|^2}{2 |S_{12}| |S_{21}|},$$

$$\Delta = (S_{11}S_{22}) - (S_{12}S_{21}),$$

$$\Gamma_s = \frac{B_1 \pm \sqrt{B_1^2 - 4|C_1|^2}}{2C_1}, \text{ where}$$

$$B_1 = 1 + |S_{11}|^2 - |S_{22}|^2 - |\Delta|^2, \text{ and}$$

$$C_1 = S_{11} - \Delta S_{22}^*,$$

$$G_T = \left(\frac{1 - \Gamma_s^2}{|1 - \Gamma_{in} \Gamma_s|^2} \right) |S_{21}|^2 \left(\frac{1 - \Gamma_L^2}{|1 - S_{22} \Gamma_L|^2} \right)$$

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Subject: - RF and Microwave Engineering (EX 752)

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- ✓ Attempt All questions.
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- ✓ Necessary chart and formula are attached herewith.
- ✓ Assume suitable data if necessary.



1. Based on operational principles, compare microwave systems with conventional low frequency systems. Lists the areas of application of microwave system. [10]
2. Illustrate different impedance matching techniques. What will be a resulting S-matrix, if you are asked to prepare a double-stub matching network of having perfect matching? [4+6]
3. Elaborate the importance of S-parameters for microwave frequencies. Evaluate the S-matrix of an Amplitude Modulator. [3+7]
4. Identify and explain the properties of a microwave passive device having following S-Matrix with necessary diagram. [10]

$$\begin{bmatrix} 0 & .0 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

5. What is cross-field effect? Describe the construction and principle of operation of a multi cavity klystron amplifier. [2+8]
6. Assuming suitable S-parameters calculate maximum gain of an amplifier working in an unilateral model. Also illustrate a flow chart showing the designing steps. [10]
7. Justify and describe how a double-section series-arm type micro strip filter is designed using insertion loss method. [10]
8. Write short notes on: (Any Two) [2×5]
 - a) Combinational effect of two magic fees connected in H-plane
 - b) VNA type Network analyzer
 - c) Microwave radiation hazards

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1. Explain the behavior of microwave circuits. Also discuss the advantages and disadvantages of using microwave frequency. [4+4]
2. Sketch a double-stub perfectly matched network using microstrip and prepare its s-matrix. [2+8]
3. For a rectangular waveguide, with suitably assumed breadth, width and frequency, (for dominant mode), determine cut-off frequency, phase velocity, propagation wavelength in the waveguide and the characteristic impedance. [6]
4. Explain the properties of two Magic Tees if one connects their E-arms and derive its S-matrix. [10]
5. Derive explicitly the field equations of Rectangular waveguides for TE modes. [8]
6. What is transit time effect? State the working principles of a magtron having phase difference of 45° of adjacent cavities. [2+6]
7. A microwave amplifier operates in Ku-Band for satellite transmit system uses a High Electron Mobility GaAs MOSFET transistor having following S-parameters at 4 GHz with 50Ω line impedance as, $S_{11} = 0.72 \angle -116^\circ$, $S_{12} = 0.03 \angle 57^\circ$, $S_{21} = 2.60 \angle 76^\circ$ and $S_{22} = 0.73 \angle -54^\circ$. Check the stability of the given transistor and compare maximum power gain for bilateral and unilateral mode. [5+5]
8. Describe in detail the procedures for prototyping Butterworth LPF using insertion loss method. [8]
9. How are microwaves hazardous to humans? Define different radiation zones of a microwave oven. [3+3]
10. Explain how microwave power is measured with static calorimeter. [6]

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 - ✓ Attempt All questions.
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 - ✓ Necessary formulas and Smith Charts are attached herewith.
 - ✓ Assume suitable data if necessary.
1. How is microwave frequency band classified by the IEEE? Enumerate the basic advantages and disadvantages of microwaves compared to lower frequencies. [5+5]
 2. A single-stub tuner is to match a lossless line to a load of an antenna. Design the stub with any assumed placement and length and derive its S-matrix. [10]
 3. A radar installation engineer is given a responsibility to install an Airport Surveillance Radar that requires half input power to the antenna than the transmitter that can deliver using a duplexer. Prepare its S-matrix. [10]
 4. Explain field and characteristics equations of a rectangular wave guide working in TM mode. [10]
 5. A microwave transistor has the following scattering parameters at 1.0GHz, with a 50Ω line impedance:
 $S_{11}=0.38\angle -158^\circ$
 $S_{12}=0.11\angle 54^\circ$
 $S_{21}=3.50\angle 80^\circ$
 $S_{22}=0.40\angle -43^\circ$
 The source impedance is $Z_s=25\Omega$ and load impedance is $Z_L=40\Omega$. Compute the gains of input and output matching networks. Also, draw a design a flowchart. [6+4]
 6. How is a low pass filter prototype based on Butterworth and Chebyshev approximations designed using insertion loss method? Implement a low pass filter double π -sections using microstrips. [8+2]
 7. Name and explain different microwave radiation fields. Describe different international EMR safety standards and practices. [5+5]
 8. Explain power measurement using static and dry calorimeters. [10]

$$K = \frac{1 - |S_{11}|^2 - |S_{22}|^2 + |\Delta|^2}{2|S_{12}S_{21}|}$$

$$\mu = \frac{1 - |S_{11}|^2}{|S_{22} - \Delta S_{11}^*| + |S_{12}S_{21}|}$$

$$C_L = \frac{(S_{22} - \Delta S_{11}^*)^*}{|S_{22}|^2 - |\Delta|^2}$$

$$R_L = \frac{|S_{12}S_{21}|}{|S_{22}|^2 - |\Delta|^2}$$

$$GT_{\max} = \frac{1}{1 - |\Gamma_S|^2} |S_{21}|^2 \frac{1 - |\Gamma_L|^2}{|1 - S_{22}\Gamma_L|^2}$$

$$\Gamma_s = \frac{B_1 \pm \sqrt{B_1^2 - 4|C_1|^2}}{2C_1}$$

$$B_1 = 1 + |S_{11}|^2 - |S_{22}|^2 - |\Delta|^2$$

$$C_1 = S_{11} - \Delta S_{22}^*$$

Exam.	Regular		
Level	BE	Full Marks	80
Programme	BEX	Pass Marks	32
Year / Part	IV / II	Time	3 hrs.

Subject: - RF and Microwave Engineering (EX752)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Necessary formulas and smith charts are provided herewith.
- ✓ Assume suitable data if necessary.

1. How the circuit at seismic band is different from its RF/Microwave counterparts? Explain. [8]
2. A 75-ohm, coaxial line is terminated with a normalized complex load of $0.4 + j0.85$ ohms. Design a double-stub matching system using short-circuited coaxial line of 75-ohm characteristic impedance. Sketch the network using micro strip. [10+2]
3. a) Analyze a three-port directional coupler using S-parameters. [4]
b) Which of the passive microwave device is explained by this S-matrix. Judge the condition and explain its characteristics.

$$[S] = \begin{bmatrix} S_{11} & 0 & S_{13} & S_{14} \\ 0 & S_{22} & -S_{13} & S_{14} \\ S_{13} & -S_{13} & 0 & 0 \\ S_{14} & S_{14} & 0 & 0 \end{bmatrix} \quad [6]$$

4. Derive the expression for the field strength for TM waves for a air-filled circular waveguide. Check the dominant mode in TE and TM modes. [8+2]
5. With neat circuit diagrams and relevant equations, explain the velocity modulation process and bunching in a multicavity reflex klystron. [10]
6. a) Refer the sketched smith chart (Fig.Q6) and analyze/synthesize the stabilities. Assume necessary parameters as desired. Mention all the steps. [5]
b) Describe the insertion loss method used for the filter designing. [5]
7. a) Discuss in detail the power measurement using circulating calorimeter. [5]
b) How microwave radiation becomes hazardous to human body? [5]
8. What do you understand by immittance chart? Sketch it. List out all duality parameters vital to designing microwave networks. [10]

Exam.	Back		
Level	BE	Full Marks	80
Programme	BEX	Pass Marks	32
Year / Part	IV / II	Time	3 hrs.

Subject: - RF and Microwave Engineering (EX752)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Necessary formula and Chart are attached herewith.
- ✓ Assume suitable data if necessary.

1. What are the advantages of using double stub matching over single stub matching? Explain the necessary steps for impedance matching of a load to a transmission line using double-stub matching network with an appropriate example. Use provided Smith Chart. [2+8]
2. Why are S-parameters used in microwave network analysis? Analyze three port network using S-parameters. [2+6]
3. Suppose there are two identical radar transmitters in equipment stock. A particular application required twice more input power to an antenna than either transmitter can deliver. Give your appropriate solution for the given problem with explanation using S-matrix. [6]
4. Describe the working principle and applications of cavity magnetron having phase shift of 45° between the cavities. [8]
5. Given S-parameters for microwave transistor amplifier: [12]

$$S_{11} = 0.78 \angle -113^\circ \quad S_{12} = 0.028 \angle 247^\circ$$

$$S_{21} = 2.60 \angle 76^\circ \quad S_{22} = 0.81 \angle -54^\circ$$

Determine the stability and compare maximum power gain for unilateral and bilateral modes using supplied formulas.
6. How is low pass filter implemented using microstrip? How are the low pass filter prototyped? [5+3]
7. What are the different types of electromagnetic radiation hazard? Illustrate radiation fields of a microwave oven. [4+4]
8. Choose a proper power measurement tool to measure power of a Airport Surveillance Radar. [8]
9. Write short notes on: [4×3]
 - i) Behavior of microwave bands
 - ii) Field equation of rectangular wave guide for TE mode
 - iii) Cavity Resonators

Exam.	Regular		
Level	BE	Full Marks	80
Programme	BEX	Pass Marks	32
Year / Part	IV / II	Time	3 hrs.

Subject: - RF and Microwave Engineering (EX752)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt **All** questions.
- ✓ The figures in the margin indicate **Full Marks**.
- ✓ **Necessary figures and Chart are attached herewith.**
- ✓ Assume suitable data if necessary.

- Design a single short and open-circuited shunt matching network for a transmission line using Smith Chart by considering an output reflection coefficient $\Gamma_L = 0.5 \angle 51^\circ$ Ohm and surge impedance $Z_0 = 50$ Ohm. [8]
- Identify and explain the properties of a microwave passive device having following S-Matrix. [8]

$$\begin{bmatrix} S_{11} & S_{12} & S_{13} & S_{14} \\ S_{21} & S_{22} & S_{23} & -S_{14} \\ S_{13} & S_{13} & 0 & 0 \\ S_{14} & -S_{14} & 0 & 0 \end{bmatrix}$$

- Sketch a flowchart for designing a microwave amplifier using a GaAsFET. Consider the following S-parameters and find maximum gain for both bilateral and unilateral model. Also using the calculated value of Γ_{in} and Γ_{out} trace Z_{in} and Z_{out} in the smith chart. [4+4+4+4]

$$[S] = \begin{bmatrix} 0.656 \angle 146.7^\circ & 0.122 \angle 46.1^\circ \\ 2.30 \angle 44.7^\circ & 0.172 \angle -117.1^\circ \end{bmatrix}$$

- Synthesize stability parameters of input matching network for the attached sketched smith chart. [8]
- Choose a proper microwave measurement tool to test an antenna as a DUT; and explain its working principles. [8]
- Explain in detail the designing steps of microwave filters. Illustrate an example of passive HPF using microstrips. [6+4]
- Express field equations of a rectangular waveguide for TM mode. [10]
- Write short notes on: (any two) [6×2]
 - Effect of SAR as microwave radiation hazards
 - Features of microwave frequency band
 - Backward Wave Oscillator
 - Microwave Cavity Resonators

Exam.	New Back (2066 & Later Batch)		
Level	BE	Full Marks	80
Programme	BEX	Pass Marks	32
Year / Part	IV / II	Time	3 hrs.

Subject: - RF and Microwave Engineering (EX 752)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Necessary Formulas and Smith Charts are attached herewith.
- ✓ Assume suitable data if necessary.

1. Differentiate between lumped and distributed circuit analysis. What are the uses of microwave bands? [4+2]
2. Assume an inductive load impedance is connected to a mismatched 50Ω transmission line. Find the size and placement of the matching stub that will remove all the standing waves and match load to the line. Use double stub shunt tuning short and open circuited stub. Draw its electrical diagram and physical connection. [8]
3. Why we use S-parameters for microwave analysis? Define S-matrix for 3 port network with appropriate example. [4+4]
4. Choose a suitable passive microwave device to split power into half and explain its properties. [8]
5. Explain what is bunching effect. Explain the working principle of BWO with neat diagrams. [2+8]
6. Check the stability and find the maximum gain a transistor amplifier having $S_{11} = 0.64\angle -169^\circ$, $S_{12} = 0.03\angle 50^\circ$, $S_{21} = 10.11\angle 91^\circ$, $S_{22} = 0.22\angle -82^\circ$. Consider both bilateral and unilateral model. Modify the S-parameters if necessary. [12]
7. Describe insertion loss method of microwave filter design. Illustrate an example of a passive LPF using μ -strip. [8+2]
8. Describe the working principle of a network analyzer. [8]
9. Write short notes on: (Any two) [2×5]
 - a) Microwave Circulators
 - b) TM mode for rectangular waveguides
 - c) Microwave radiation hazards

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Exam.	Regular		
Level	BE	Full Marks	80
Programme	BEX	Pass Marks	32
Year / Part	IV / II	Time	3 hrs.

Subject: - RF and Microwave Engineering (EX752)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Necessary formula, graph and figures are attached herewith.
- ✓ Assume suitable data if necessary.

1. Classify signal frequency in different bands of waves and rays. What are the advantages and disadvantages of using microwave signal? [3+5]
2. By assuming a complex inductive load of an antenna which is mismatched with the line impedance of 78.0 ohm, design a double-stub short-circuited matching network. Show both electrical and physical connections. [8+2]
3. Why S-parameter is important in microwave network analysis? Write down the properties of a 3-port network. [4+4]
4. Suppose there are two identical radar transmitters and few passive devices in equipment stock. A particular application requires twice more input power to an antenna than either transmitter can deliver. As a RF engineer, give your appropriate solution for the above problem with necessary figures, mathematics and sufficient explanation. [8]
5. What do you mean by slow backward wave structure? Explain the construction and working principle of a LNA. [2+6]
6. Show a flow diagram that explain designing of an amplifier using a FET transistor. With self-defined parameters and the help of a smith chart define conditional stability of a microwave amplifier. [10]
7. Justify and describe how a microwave filter is designed using insertion loss method. [2+6]
8. Define major microwave measurement parameters and explain the working principle of a low microwave power measurement device. [8]
9. Write short notes on: (any two) [2×6]
 - a) RF/MW radiation hazards and safety practices
 - b) Directional Couplers
 - c) TE mode circular wave guide

Exam.	New Back (2066 & Later Batch)		
Level	BE	Full Marks	80
Programme	BEX	Pass Marks	32
Year / Part	IV / II	Time	3 hrs.

Subject: - RF and Microwave Engineering (EX752)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Necessary Formulas and Smith Charts are attached herewith.
- ✓ Assume suitable data if necessary.

1. What are the merits and demerits of microwaves compared to seismic wave? [5]
2. Compare RF radiation fields and explain how they are hazardous to living body. [5]
3. Sketch an immittance chart and compare the scales. [5]
4. Design a signal shunt and open matching networks using Smith Chart for a transmission line having surge impedance of 75 Ohm and load impedance of $78.27 + j60.93$ Ohm. Sketch the physical diagram considering microstrips. [4+4+2]
5. You are supposed to measure about 7.5 mW of microwave power. Choose a proper power measuring device and explain its working principle. [10]
6. What is cross-field effect? Design and describe the working principle of a cross field cavity device for power amplifier. Consider a 90° phase shift between adjacent cavities. [3+7]
7. Describe magic Tee based on S-parameters. Differentiate between dominant and degenerate modes. Consider a rectangular waveguide having dimension of $a = 3b$ and find the dominant mode among TM_{01} , TM_{10} , TM_{11} , TM_{21} , TM_{12} , TM_{02} and TM_{20} . [6+2+2]
8. A GaAs FET transistor has the S-Parameters at 5 GHz with 50 Ohm line measured as $S_{11} = 0.45\angle 163^\circ$, $S_{12} = 0.04\angle 40^\circ$, $S_{21} = 2.55\angle -106^\circ$ and $S_{22} = 0.46\angle -65^\circ$. Check the stability and find the maximum power gain using this transistor. [10]
9. Write short note on: (any three) [3×5]
 - a) Microwave filter parameters and LPF prototyping
 - b) TM mode rectangular wave guide
 - c) Bunching effect in reflex klystron
 - d) Microwave mixer

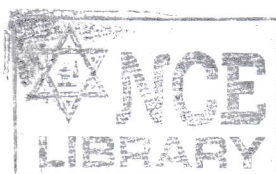


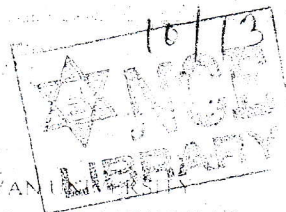
Exam.	Regular		
Level	BE	Full Marks	80
Programme	BEX	Pass Marks	32
Year / Part	IV / II	Time	3 hrs.

Subject: - RF and Microwave Engineering (EX752)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Necessary Formulas and Smith Charts are attached herewith.
- ✓ Assume suitable data if necessary.

1. What are the advantages and disadvantages of microwaves over acoustic waves? [5]
2. What is admittance chart? A load impedance of $Z_L = 80 + j100$ is connected to a microstrip transmission line. Find the size and placement of the matching stub. Use single stub shunt tuning short and open stubs. [2+8]
3. Define the use of S-parameters for three-port analysis. Define the term return loss and insertion loss. [5+2]
4. What are waveguide junctions? Describe the operational principles of magic tee based on s-parameters. [3+3]
5. What is density modulation? Describe the working principle of a multi-cavity klystron oscillator. [2+7]
6. Justify that a transistor having following S-parameters $S_{11} = 0.894 \angle -60.6^\circ$, $S_{12} = 0.020 \angle 62.4^\circ$, $S_{21} = 3.122 \angle 123.6^\circ$ and $S_{22} = 0.781 \angle -27.6^\circ$ is conditionally stable while designing an amplifier. Considering unilateral model calculate maximum gain. [5+5]
7. How can you implement low pass filter using micro-strip? How they are prototyped? [3+5]
8. Describe how standing waves and microwave powers are measured with VSWR meter and low power measurement. [2+8]
9. Write short notes on: (any three) [3×5]
 - a) Dominant mode in waveguide
 - b) Circulators
 - c) LNA cavity device inserting loss method for filter designing
 - d) Insertion loss method for filter designing





Exam.	New Back (2066 & Later Batch)		
Level	BE	Full Marks	80
Programme	BEX	Pass Marks	32
Year / Part	IV / II	Time	3 hrs.

Subject - RF and Microwave Engineering (EX752)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Necessary figures and Smith Charts are attached herewith.
- ✓ Assume suitable data if necessary.

1. Differentiate the behaviors of the systems at microwave and conventional low frequency bands. [6]
2. Describe how TE mode is different from TM mode in a circular waveguide. [10]
3. Describe the working principle of a cavity magnetron. [10]
4. Why S-parameter is important in microwave network analysis? Define S-parameters for a two-port network. [4+5]
5. By arbitrarily assuming a suitable load that connects to a 50-ohm transmission line find the lengths and spacing for a two-stub impedance matching system. Assume also a suitable separation between the stubs. [10]
6. Using the following S-parameters of $S_{11}=0.55\angle-150^\circ$, $S_{12}=0.04\angle20^\circ$, $S_{21}=2.82\angle180^\circ$ and $S_{22}=0.45\angle-30^\circ$, calculate and compare maximum power gain for unilateral and bilateral modes. [15]
7. Discuss the difference between an amplifier circuit and an oscillator circuit in terms of stability factor. [5]
8. Write short notes (Any THREE) [3 x 5]
 - a. Microwave magic tee
 - b. Microwave radiation fields
 - c. Microwave strip-lines against micro-strips
 - d. Static calorimeter

Exam.	Regular / Back		
Level	BE	Full Marks	80
Programme	BEX	Pass Marks	32
Year / Part	IV / II	Time	3 hrs.

Subject: - RF and Microwave Engineering (EX752)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Necessary formulas and Smith Charts are attached herewith.
- ✓ Assume suitable data if necessary.

- Describe field equations and other related parameters of a rectangular waveguide in TM mode. Compare TE₁₀ and TE₂₀ in terms of cut-off frequency and dominant mode. [8+2]
- Design a double stub matching network using three-eighths wavelength $\left(\frac{3\lambda}{8}\right)$ separation that match an antenna having load of 300+j300 Ohm connected to a 300 Ohm transmission line. Justify your design. [10]
- What is bunching effect? Describe the working principle of a klystron oscillator. [2+8]
- Using the given S-parameters $S_{11}=0.55\angle 150^\circ, S_{12}=0.04\angle 20^\circ, S_{21}=2.82\angle 180^\circ, S_{22}=0.45\angle -30^\circ$ and required assumptions, calculate maximum gains of this transistor amplifier for bilateral and unilateral modes. [10]
- Draw a flow diagram to describe the design procedure of a microwave amplifier. Define the stability of an amplifier having $C_S=1.15\angle 10^\circ, R_S=0.85, C_L=1.10\angle 80^\circ, R_L=1.10$. [5+5]
- How microwave measurements are different to low frequency measurements? Describe how static calorimeter works to measure power. [3+7]
- Design a two-port network model and derive the required parameters. [10]
- Write short notes (Any TWO) [2x5]
 - Design procedures of microwave filters
 - Microwave radiation hazards and safety practices
 - Backward Wave Oscillator
 - Merits of S-parameters in microwaves

Supplied Formulas:

$$K = \frac{1 + |\Delta|^2 - |S_{11}|^2 - |S_{22}|^2}{2|S_{12}||S_{21}|}$$

$$\Delta = (S_{11}S_{22}) - (S_{12}S_{21})$$

$$\mu = \frac{1 - |S_{11}|^2}{|S_{22} - \Delta S_{11}^*| + |S_{21}S_{12}|}$$

$$\Gamma_S = \frac{B_1 \pm \sqrt{B_1^2 - 4|C_1|^2}}{2C_1}$$

$$\Gamma_L = \frac{B_2 \pm \sqrt{B_2^2 - 4|C_2|^2}}{2C_2}, \text{ where}$$

$$B_1 = 1 + |S_{11}|^2 - |S_{22}|^2 - |\Delta|^2$$

$$B_2 = 1 + |S_{22}|^2 - |S_{11}|^2 - |\Delta|^2$$

$$C_1 = S_{11} - \Delta S_{22}^*, \text{ and}$$

$$C_2 = S_{22} - \Delta S_{11}^*$$

$$C_L = \frac{(S_{22} - \Delta S_{11}^*)}{|S_{22}|^2 - |\Delta|^2}$$

$$C_S = \frac{(S_{11} - \Delta S_{22}^*)}{|S_{11}|^2 - |\Delta|^2}$$

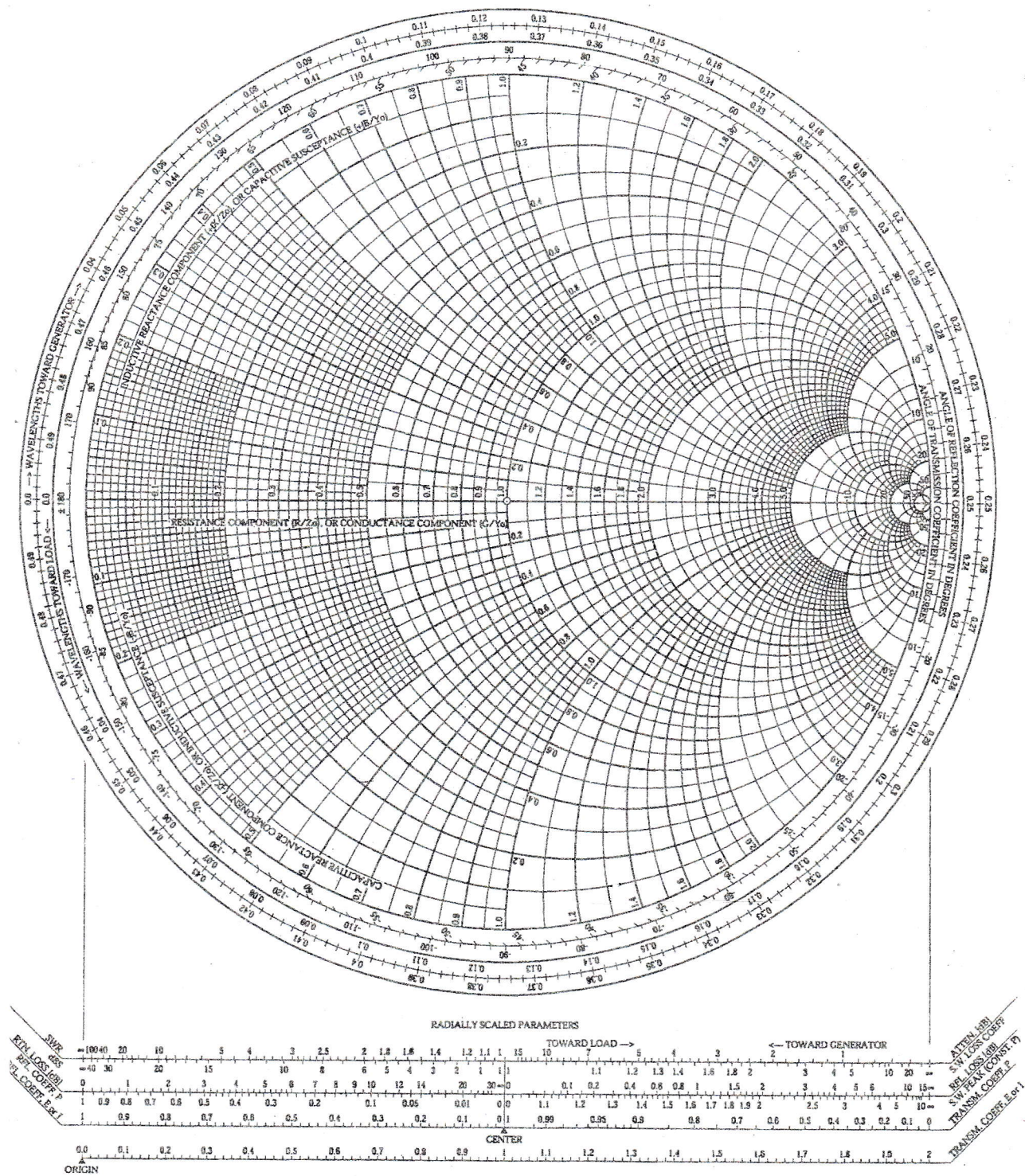
$$R_L = \frac{|S_{12}S_{21}|}{|S_{22}|^2 - |\Delta|^2}$$

$$R_S = \frac{|S_{12}S_{21}|}{|S_{11}|^2 - |\Delta|^2}$$

$$G_{max} = \left(\frac{1 - |\Gamma_S|^2}{|1 - S_{11}\Gamma_S|^2} \right) |S_{21}|^2 \left(\frac{1 - |\Gamma_L|^2}{|1 - S_{22}\Gamma_L|^2} \right)$$

The Complete Smith Chart

Black Magic Design



Exam.	Regular		
Level	BE	Full Marks	80
Programme	BEX	Pass Marks	32
Year / Part	IV / II	Time	3 hrs.

Subject: - RF and Microwave Engineering (EX752)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Necessary formulas and Smith charts are attached herewith.
- ✓ Assume suitable data if necessary.

- Based on operational principles, compare microwave systems with conventional low frequency systems. Lists the areas of application of microwave systems. [4+4]
- What makes S-parameters useful in microwave network analysis? Define S-parameters for a two-port network. Justify that the Butterworth and Chebyshev filter responses are common to prototype microwave two-port filter network using insertion loss method. [4+4+4]
- Design a double-stub impedance matching network for a given load of $80 + j180$ Ohm connected to a 100-Ohm transmission line at 3 GHz with a three-eighths wavelength separation between the stubs. Illustrate necessary diagrams to show physical connections. [8+2]
- Define expressions for various field components of a rectangular waveguide in TE mode. Show that a 1 GHz signal cannot propagate in TE₁₀ made in a rectangular waveguide with a wall separation of 5 cm. [7+3]
- Find the maximum gain for a microwave transistor amplifier with $S_{11} = 0.656 \angle 146.7^\circ$, $S_{12} = 0.122 \angle 46.1^\circ$, $S_{21} = 2.3 \angle 44.7^\circ$, $S_{22} = 0.172 \angle -117.1^\circ$. [10]
- What is bunching effect? Briefly describe the construction and operational features of a cavity magnetron. [2+8]
- Describe how standing waves and microwave powers are measured with VSWR meter and bolometry respectively. [4+6]
- Write short notes on: (any two) [2×5]
 - Mixer theory
 - Circulators
 - Microwave radiation hazards and safety practices

Supplied Formulas:

$$K = \frac{1 + |\Delta|^2 - |S_{11}|^2 - |S_{22}|^2}{2|S_{12}| |S_{21}|},$$

$$\Delta = (S_{11}S_{22}) - (S_{12}S_{21}),$$

$$\mu = \frac{1 - |S_{11}|^2}{|S_{22} - \Delta S_{11}^*| + |S_{21}S_{12}^*|},$$

$$\Gamma_s = \frac{B_1 \pm \sqrt{B_1^2 - 4|C_1|^2}}{2C_1},$$

$$\Gamma_L = \frac{B_2 \pm \sqrt{B_2^2 - 4|C_2|^2}}{2C_2}, \text{ where}$$

$$B_1 = 1 + |S_{11}|^2 - |S_{22}|^2 - |\Delta|^2,$$

$$B_2 = 1 + |S_{22}|^2 - |S_{11}|^2 - |\Delta|^2,$$

$$C_1 = S_{11} - \Delta S_{22}^*, \text{ and}$$

$$C_2 = S_{22} - \Delta S_{11}^*,$$

$$C_L = \frac{(S_{22} - \Delta S_{11}^*)}{|S_{22}|^2 - |\Delta|^2},$$

$$C_s = \frac{(S_{11} - \Delta S_{22}^*)}{|S_{11}|^2 - |\Delta|^2},$$

$$R_L = \frac{|S_{12}S_{21}|}{|S_{22}|^2 - |\Delta|^2},$$

$$R_s = \frac{|S_{12}S_{21}|}{|S_{11}|^2 - |\Delta|^2},$$

$$G_{r_{\max}} = \left(\frac{1}{1 - |\Gamma_s|^2} \right) |S_{21}|^2 \left(\frac{1 - |\Gamma_L|^2}{|1 - S_{22}\Gamma_L|^2} \right)$$

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Exam.	Regular / Back		
Level	BE	Full Marks	80
Programme	BEX	Pass Marks	32
Year / Part	IV / II	Time	3 hrs.

Subject: - RF & Microwave Circuits, Systems & Devices (EG785EX) (Elective II)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Necessary formulas and smith chart are attached herewith.
- ✓ Assume suitable data if necessary.

1. Classify microwave frequency bands and state their major applications. Describe how microwave transmission lines are different from the conventional low frequency transmission lines. [4+4]
2. Describe microwave radiation hazards based on the radiation fields. [8]
3. Describe rectangular waveguide based on modes on propagation and other critical parameters. [8]
4. Describe why S-parameter is important in microwave network analysis. Using a two-port network derive S-parameters. [4+6]
5. What is double-stub tuner? Assuming a load of $75 + j75$ ohm is connected to a 50-ohm transmission line, find the lengths and spacing for a two-stub impedance matching system with three-eighths wavelength separation between the stubs. [3+15]
6. Design an amplifier to attain maximum gain at 4.0 GHz using a GaAs FET having following S-parameters: $S_{11} = 0.72\angle -116^\circ$, $S_{12} = 0.03\angle 57^\circ$, $S_{21} = 2.60\angle 76^\circ$ and $S_{22} = 0.73\angle -54^\circ$. Consider the characteristic impedance, $Z_0 = 50$ Ohm. [18]
7. Write short notes (Any TWO)
 - a. E-plane tee against H-plane tee
 - b. PROBE-coupling against LOOP-coupling
 - c. Microstrips
 - d. Two-cavity klystron

[2 x 5]
